

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– VI (NEW) EXAMINATION – WINTER 2021****Subject Code:3160918****Date:02/12/2021****Subject Name:Element of Electrical Design****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

- Q.1** (a) Classify insulating materials on the basis of maximum permissible temperature rise. **03**
- (b) Derive the equation of gap contraction factor for slots. **04**
- (c) Describe design steps for small single phase transformer. **07**
- Q.2** (a) Define the following terms used in armature winding design: **03**
(1) Coil span, (2) Pole pitch (3) front pitch
- (b) Describe the field control method and its application for DC shunt generator and DC shunt motor. **04**
- (c) Design a mush winding for 3-phase, 4-pole and 24 slots stator. Also show winding diagram for R phase. **07**

OR

- (c) Calculate the Front pitch, back pitch, winding pitch and Commutator pitch for a simplex lap wound 16 slots, 4-pole DC armature. Make the winding table and draw the winding diagram in developed form. Also draw the sequence diagram to indicate the position of brushes. Assume number of coil sides per slot = 2. **07**
- Q.3** (a) Write steps to estimate the total cost of electrical wiring installation for building. **03**
- (b) Justify the use of dummy coils and equalizer connections in DC armature windings. **04**
- (c) Determine the MMF required for the airgap of a machine with open slots from the following particulars: **07**
Slot pitch= 4.3 cm, Slot opening= 2.1 cm, Gross length of core= 48 cm, Pole arc= 18 cm, Airgap length= 0.6 cm, Flux per pole= 0.056 wb, No. of ventilating ducts= 8, Width of each ventilating ducts= 1.2 cm

Slot opening/Gap length	1	2	3	3.4	4
Carter's coefficient	0.15	0.28	0.37	0.41	0.43

OR

- Q.3** (a) Discuss the necessity of starter in DC motors and Induction motors. **03**
- (b) Find whether the following wave windings are symmetrical or not: **04**
(i) 4-pole, 25 slots, 2 coil sides/slot
(ii) 4-pole, 14 slots, 2 coil sides/slot
- (c) Define real and apparent flux densities in the tooth of DC machine armature and give the difference between them. Also derive the relation between them. **07**
- Q.4** (a) Describe how to calculate the magnetizing current in a machine with distributed winding. **03**
- (b) Compare Simplex lap winding & Simplex wave winding. **04**

- (c) Explain the grading of starting resistance for DC Series motor starters. **07**
- OR**
- Q.4** (a) Explain working of No Volt Coil and Overload Release coil in terms of three point starters for DC motor. **03**
- (b) Discuss the factors considered for selection of type of electrical wiring. **04**
- (c) Design a suitable 8 section starter for a 14.92 kw, 250 V, 1000 rpm D.C. shunt motor from the following data:
Maximum starting torque = Full load torque, Armature circuit resistance = 0.4 ohm, Full load efficiency = 0.85 **07**
- Q.5** (a) State and explain methods for calculating MMF required for tooth in DC machine. **03**
- (b) Explain the working of auto transformer starter with neat sketch for squirrel cage induction motor. **04**
- (c) Give the design steps for three phase variable chock coil. **07**
- OR**
- Q.5** (a) Discuss the significance of (i) Space to height ratio (ii) Utilization factor and (iii) Depreciation factor in the design of lighting system. **03**
- (b) Discuss briefly the different types of loads with examples. **04**
- (c) A small room of size 4 m X 3 m is required to be provided with lamp, fan, tube light and one 5A 3-pin socket outlet. Each of the points is controlled with their respective switches installed in one switch board. Assume PVC wiring system. No main switch is to be provided as the entry of the sub-circuit is from nearby room. **07**
- Do the following:
- (i) Mark the location of electrical points and draw the installation plan.
- (ii) Draw the wiring and schematic diagram.
- (iii) Calculate the length of phase and neutral wire required for the winding installation.
- (iv) Calculate the length of earth wire required.
